

Guy Azran – CV

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Research Interests

Sequential decision-making – state estimation, planning under uncertainty, task and motion planning, reinforcement learning.

Deep learning – learned abstractions, computer vision, multimodal foundation models.

Classical machine learning – conformal prediction, multilabel classification.

Education

- 2023 – present: PhD candidate in computer science, Technion – Israel Institute of Technology
 - Research focus: State estimation for reliable autonomy under perceptual uncertainty in robot systems.
- 2020 – 2023: MSc in Computer Science (Thesis track), Technion – Israel Institute of Technology
 - Thesis: *Contextual Pre-Planning on Reward Machine Abstractions for Enhanced Transfer in Deep Reinforcement Learning*, published as an oral paper at AAAI 2024.
 - GPA 96.3%
- 2016 – 2020: BSc in Computer Science and Mathematics, Hebrew University of Jerusalem
 - Dual degree with a strong mathematical foundation (analysis, algebra, probability, logic).
 - Undergraduate research on automated analysis of microscopy data (arXiv:2009.12684).
 - GPA of 91.9%; Dean's list (first year)

Work Experience

- June 2025 – September 2025: Summer internship at IBM research working on LLM inference-time error correction
- November 2022 – March 2025: Student researcher, Intel Labs. Autonomous Agents/Extended Realities Lab
- August 2017 – October 2020: Student Wi-Fi firmware developer, Intel.
- March 2015 - November 2016: Independent website and mobile application developer.

Teaching Experience

- ICAPS 2024 summer school: Teacher and co-creator of labs.
- Supervised multiple undergraduate projects

- Teaching assistant in undergraduate and graduate level courses.
 - AI and Robotics (Head TA)
 - Planning and Reinforcement Learning in AI systems (Head TA)
 - From Single Agent to Multi-Agent Sequential Decision Making Under Uncertainty (Head TA)
 - Introduction to Machine Learning
 - Introduction to Computer Science

Publications

Conferences

- **Guy Azran**, Sarah Keren. **Robust Task-Aware State Estimation from Pretrained Perception**. The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026). Also presented at the IROS Workshop on Rethinking Uncertainty for Modern Robotics Paradigms 2026 (oral presentation).
- **Guy Azran**, Mohamad Hosein Danesh, Stefano V. Albrecht, Sarah Keren. **Contextual Pre-planning on Reward Machine Abstractions for Enhanced Transfer in Deep Reinforcement Learning**. Proceedings of the AAAI Conference on Artificial Intelligence 2024 (oral presentation).
- Ilai Shraga, **Guy Azran**, Matthias Gerstgrasser, Ofir Abu, Jeffrey S. Rosenschein, Sarah Keren. **Collaboration Promotes Group Resilience in Multi-Agent RL**. Reinforcement Learning Conference 2025 (oral presentation).
- **Guy Azran**, Michael Navat, Sarah Keren. **Bridging Learned Visual Perception and Symbolic Belief-Space Planning via Probabilistic Grounding**. International Conference on Neuro-Symbolic Systems (NeuS) 2026 (oral presentation). Also presented at the IROS 2026 Workshop on Neuro-Symbolic World Models (RoBoWoMo) (spotlight and 3rd place best paper).

Workshops

- **Guy Azran**, Mohamad Hosein Danesh, Stefano Albrecht and Sarah Keren. **Enhancing Transfer of Reinforcement Learning Agents with Abstract Contextual Embeddings**. NeurIPS Workshop on Neuro Causal and Symbolic AI 2022.
- **Guy Azran**, Yuval Goshen, Kai Yuan, Sarah Keren. **S3E: Semantic Symbolic State Estimation With Vision-Language Foundation Models**. AAAI Workshop on Planning in the Era of LLMs 2025.
- Eliezer Shlomi, **Guy Azran**, Eilam Shapira, Omer Nahum, Guy Uziel, Michael Katz, Ateret Anaby Tavor, Roi Reichart, Sarah Keren. **Transition Function Prediction in AI Planning Using LLMs**. AAAI Workshop on Planning in the Era of LLMs 2025 and ICRA Workshop on Language and Semantics of Task and Motion Planning.
- Matan Argaman, **Guy Azran**, Sarah Keren. **Factored State Sampling**. AAAI Workshop on Planning in the Era of LLMs 2025.

- **Guy Azran**, David Dovrat, Sarah Keren. **Task-Aware Conformal State Estimation for Reliable Robotic Planning**. IROS Workshop on Embodied Neuro-Symbolic AI for Reliable and Safe Robotics (ReS AI) 2026 (oral presentation).
- **Guy Azran**, Sarah Keren. **Minimal Particle Set Tracking with Neurosymbolic State Estimation for Robust Planning and Execution**. IROS Workshop on Neuro-Symbolic AI for Intelligent Robots (NSAIR) 2026 (spotlight)

Preprint

- Jonathan Reiner, Guy Azran, Gal Hyams. MicroAnalyzer: **A Python Tool for Automated Bacterial Analysis with Fluorescence Microscopy**. arXiv:2009.12684.

Presentations:

- Guy Azran, David Dovrat, Sarah Keren. **Task-Aware Conformal State Estimation for Reliable Robotic Planning**. Israeli Conference of Mechanical Engineering 2026 (oral presentation).

Open-source contributions:

- Co-creator and maintainer of ICAPS 2024 summer school tutorials on task and motion planning.
- Contributor to Gymnasium (previously OpenAI/gym).
- Contributor to the Unified Planning Framework
- Main author and maintainer of multi-taxi environment.
- Main author and maintainer of the S3E package.
- Co-author and maintainer of a series of tutorials on task and motion planning.

Honors & Awards

- 3rd Place Best Paper Award, RoBoWoMo @ IROS 2026
- Outstanding TA (2026)
- Spontaneous Recognition Award, Intel (2019).
- Dean's List, Hebrew University of Jerusalem (2017).

Professional Service

- Co-organizer of the Workshop on Reliability in Planning and Learning (RIPL) at ICAPS 2026.
- Co-creator of the ICAPS 2024 summer school
- Reviewer, 2024–present: AAAI, NeurIPS, ICML, ICLR, AAMAS, ICAPS, CoRL, NeuS

Technical Skills

Machine Learning & AI

- Deep Learning Frameworks: PyTorch, TensorFlow, Keras
- ML Libraries: scikit-learn, SciPy, Hugging Face Transformers
- Simulation & Robotics: MuJoCo physics simulator, ROS2

Laboratory Experience

- Robotics: UR5 experimentation
- Hardware: Wi-Fi test environment construction and debugging
- Data Collection & Analysis: Automated microscopy data analysis

Programming Languages

- Primary: Python (advanced)
- Additional: C/C++, Java, C#, web (HTML, CSS, JavaScript)

Development & Research Tools

- Version Control: Git & GitHub (some experience with Perforce P4V)
- Development Environments: Jupyter Notebooks, Anaconda
- System: Bash/PowerShell/Batch scripting, High-performance computing (SLURM)
- LLM-driven coding: Claude code and OpenAI Codex

Military Service

- 2010 - 2015: IDF Mandatory and officer service.
 - Served in combat field intelligence as a combat officer and platoon leader.
 - Honorably discharged at the rank of Lieutenant.
- 2015 - present: IDF Reserve duty:
 - Served in homefront command and other intelligence units.
 - Current rank: Captain.

Personal Background

- Born in Israel. Lived in Asia and Europe for 8 years between ages 6-14.
- Attended international schools in English; culturally fluent and adaptable.